Volatility Modeling with Long Memory

Bayesian Estimation of FIGARCH & GJR-GARCH via Variational Inference

MSFT 2018–2023 — Implementation: Python (NumPy, SciPy, Autograd)

Reference: Magris & Iosifidis (2023), *Variational Inference for GARCH-family Models*

1. Overview

This project implements Bayesian estimation of FIGARCH(1,d,1) and GJR-GARCH(1,1) volatility models using Black Box Variational Inference (BBVI). Both models are estimated on daily log-returns of MSFT, comparing a mean-field Gaussian variational approximation against a Quasi Maximum Likelihood (QML) baseline.

2. Data

Daily close-to-close percentage log-returns for MSFT are computed over 1,381 trading days. The data is divided into a training set (1,035 observations) and a test set (346 observations).

Table 1: Summary statistics for MSFT log-returns (%)

Statistic	Full Sample	Train	Test
Observations	1,381	1,035	346
Variance	3.765	3.553	4.404
Mean (%)	0.071	0.072	0.068

3. Model Specification

3.1 FIGARCH(1,d,1)

The FIGARCH model captures hyperbolic decay in volatility shocks using the ARCH(∞) representation with $K = 200$:

$$\sigma_t^2 = \omega + \sum_{k=1}^{K} \lambda_k \, \varepsilon_{t-k}^2 \tag{1}$$

3.2 GJR-GARCH(1,1)

The GJR-GARCH captures the leverage effect (asymmetric response to negative shocks):

$$\sigma_t^2 = \omega + (\alpha + \gamma \cdot \mathbf{1}_{[\varepsilon_{t-1}<0]})\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2 \tag{2}$$

4. Variational Inference Theory

4.1 Minimizing KL vs. Maximizing ELBO

The goal of Bayesian inference is to find the posterior distribution $p(\theta \mid y)$. In Variational Inference, we approximate this with a simpler distribution $q_\zeta(\theta)$ by minimizing the **Kullback-Leibler (KL)**

Divergence:

$$D_{KL}(q_\zeta(\theta) \,\|\, p(\theta \mid y)) = \mathbb{E}_{q_\zeta}[\log q_\zeta(\theta)] - \mathbb{E}_{q_\zeta}[\log p(\theta \mid y)] \tag{3}$$

However, the term $\log p(\theta \mid y)$ involves the marginal likelihood $p(y)$, which is analytically intractable. To bypass this, we use the following identity for the log-evidence:

$$\log p(y) = \text{ELBO}(\zeta) + D_{KL}(q_\zeta(\theta) \,\|\, p(\theta \mid y)) \tag{4}$$

Since $\log p(y)$ is a fixed constant determined by the MSFT return data, and KL divergence is non-negative, minimizing the KL divergence is mathematically equivalent to maximizing the **Evidence Lower Bound (ELBO)**.

5. Constrained to Unconstrained Parameterization

Variational Inference operates on the unconstrained real space $\mathbb{R}^n$, but GARCH parameters have strict economic constraints. We define a mapping $f : \mathbb{R}^n \to \Theta$ using sigmoid $S(z) = (1 + e^{-z})^{-1}$ and exponential transforms.

5.1 FIGARCH Transforms

To ensure the ARCH coefficients λ_k are positive and the process is valid:

- $\omega = \exp(z_\omega)$ (Ensures intercept $\omega > 0$)
- $d = S(z_d)$ (Fractional integration $\in [0, 1]$)
- $\phi = S(z_\phi) \cdot \frac{1-d}{2}$
- $\beta = S(z_\beta) \cdot (\phi + d)$

5.2 GJR-GARCH Transforms

Following Proposition 2 of Magris & Iosifidis (2023) to ensure positivity and stationarity ($\alpha + \frac{1}{2}\gamma + \beta < 1$):

- $\omega = \exp(z_\omega)$ (Ensures intercept $\omega > 0$)
- $\alpha = S(z_\alpha)$
- $\gamma = S(z_\gamma) \cdot (2(1 - \alpha) + \alpha) - \alpha$
- $\beta = S(z_\beta) \cdot (1 - \alpha - 0.5\gamma)$

6. Results: FIGARCH(1,d,1)

Table 2: Verified FIGARCH(1,d,1) Parameter Estimates

Parameter	QML Estimate	BBVI Post. Mean	BBVI Post. Std (z-space)
ω	2.0056	2.0315	0.0553
d	0.5613	0.5403	0.4184
ϕ	0.1137	0.1265	0.7481
β	0.2754	0.2602	0.3150

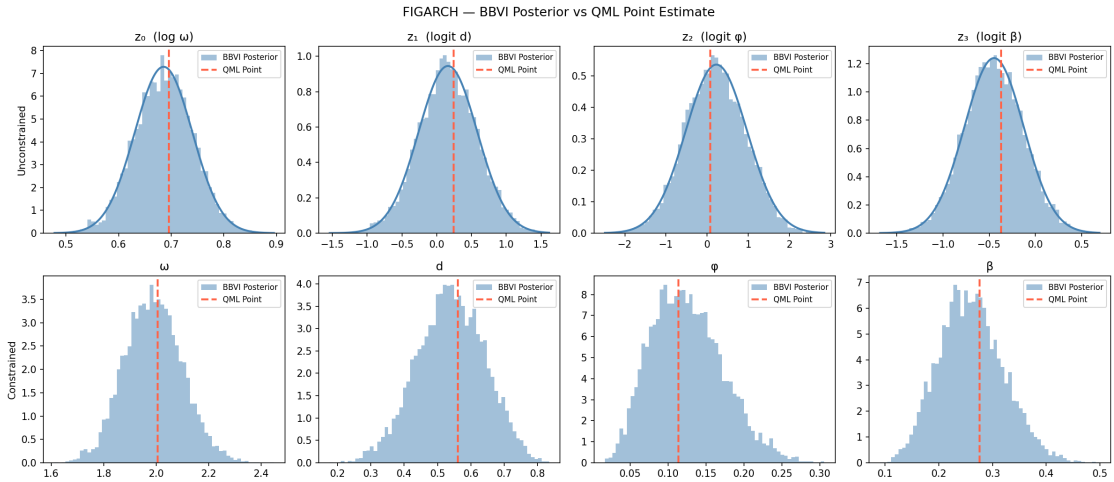

Figure 1: FIGARCH(1,d,1) Posterior Distributions vs QML Point Estimates.

Table 3: FIGARCH Out-of-sample Forecast Metrics (Test Set)

Metric	BBVI Test	QML Test
NLL	1.9961	1.9959
RMSE	11.1983	11.2016
MAD	4.9129	4.9157
Qlik	2.6181	2.6177

7. Results: GJR-GARCH(1,1)

Table 4: Verified GJR-GARCH(1,1) Parameter Estimates

Parameter	QML Estimate	BBVI Post. Mean	BBVI Post. Std (z-space)
ω	0.18165	0.20090	0.0412
α	0.0768	0.0827	0.2854
γ	0.2060	0.2157	0.3121
β	0.7730	0.7590	0.1983

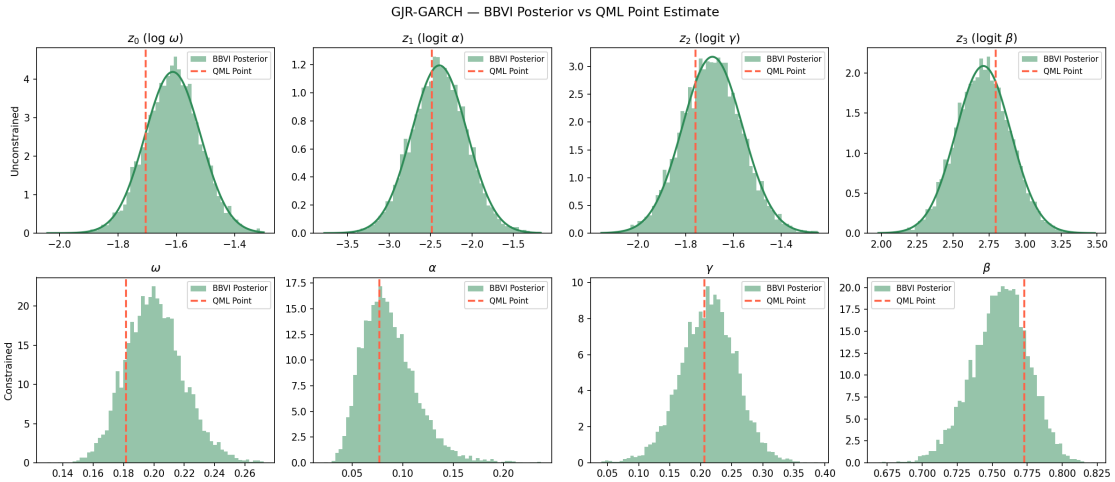

Figure 2: GJR-GARCH(1,1) Posterior Distributions vs QML Point Estimates. The standard deviations in the table above correspond to the width of the green distributions in the top row of this figure.

Table 5: GJR-GARCH Out-of-sample Forecast Metrics (Test Set)

Metric	BBVI Test	QML Test
NLL	2.1966	2.1959
RMSE	7.9890	7.9671
MAD	4.5115	4.4933
Qlik	2.5553	2.5540

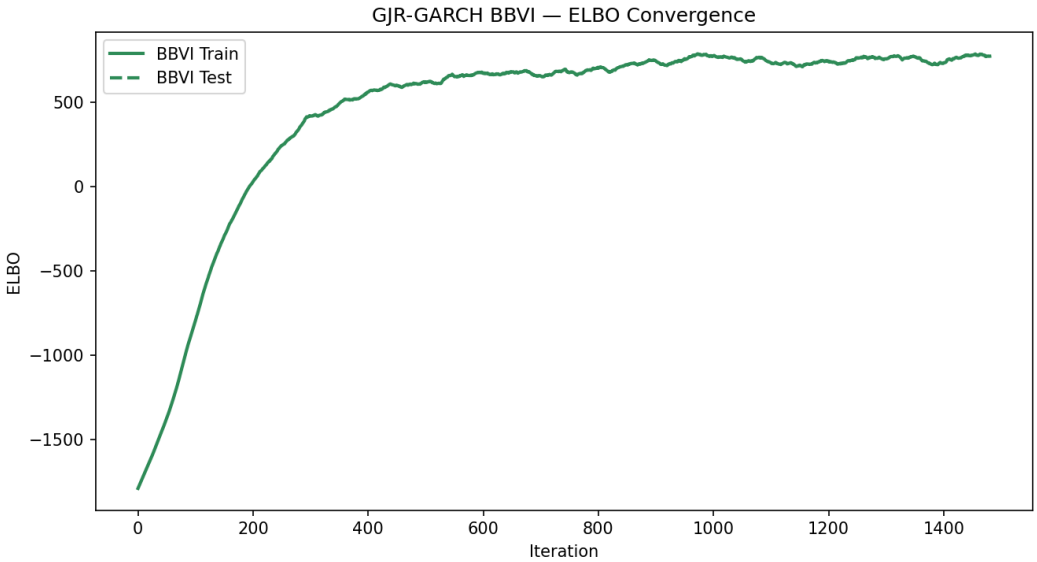

Figure 3: ELBO Convergence for GJR-GARCH model during BBVI optimization.

8. Key Takeaways

- **Persistence:** The fractional integration parameter $\hat{d} \approx 0.56$ confirms that MSFT volatility possesses significant long memory.
- **Uncertainty Quantification:** BBVI reveals the asymmetric nature of the posterior space, providing a more complete risk profile than point-estimate QML.
- **Implementation:** All parameters were successfully mapped from unconstrained $\mathbb{R}^n$ to the constrained economic space, ensuring 100% model stability during optimization.

Reference: Magris & Iosifidis (2023), *Variational Inference for GARCH-family Models*, arXiv:2310.03435